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Face Recognition Detection Based on PCA-SVM

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***Abstract*—In order to improve the application efficiency of face recognition technology, this paper proposes a face recognition model based on PCA-SVM. Firstly, the basic ORL face database is updated to build a new face recognition database; secondly, the PCA algorithm is used to reduce the dimensionality of face features, and the dimensionality-reduced face features are passed into the SVM classifier for training; finally, based on The PCA-SVM algorithm establishes a face recognition model, and compares the recognition effect of the PCA model and the improved PCA-SVM model through examples to verify the reliability and accuracy of the model.**

Keywords- face recognition; MATLAB; PCA dimensionality reduction; SVM

I. INTRODUCTION

With the overall development of science and technology, the application of face recognition is becoming more and more extensive. Many experts at home and abroad have explored it. Sun Degang [1] analyzed the PCA algorithm theory, combined with face recognition The basic idea and process put forward the application principle of the algorithm in face recognition, and discussed the implementation method of the algorithm; Orison [2] used the SVM multi-class classifier to find the optimal segmentation hyperplane between face samples, The characteristic face was trained and classified.

Face recognition technology is based on the basic biological characteristics of a creature, identifying it, so as to make a judgment on the identity of the identified organism. Different people's face information is not exactly the same, and face recognition technology is based on such a premise, through the identification of different face information, the identity of different people is recognized and judged. Many of the features of facial recognition technology are not unique, and these characteristics are also reflected in other technologies. However, facial recognition technology also has its own characteristics, such as visual characteristics: the ability to recognize faces by simulating visual senses, as well as the characteristics of being easy to use, not easy to detect, and clear results.

Face recognition technology consists of four sets of processing processes, which are face image acquisition and detection, face image preprocessing, face image feature extraction, and face image matching and recognition . The main function of face image acquisition and detection is to detect the face image uploaded by the user and collect information at the same time. Face image preprocessing is to pre-process the image uploaded by the user, so that the features of the face image can be extracted efficiently in the later stage. This kind of preprocessing is necessary because user-uploaded images are often unusable for a variety of reasons, and the preprocessing process mainly includes light compensation, normalization, geometric correction, facial image filtering, and sharpening . The facial attribute extraction process, also known as the facial character model, mainly models the facial model with the aim of extracting better facial features; Facial feature extraction can also be referred to as face feature extraction, the process is mainly for some facial features, through facial modeling, in order to better extract facial features: facial recognition is to compare facial features, recognize the obtained facial feature format, and judge facial identity information based on similarity. This process is further divided into two categories: one is confirmation, which is the process of one-to-one image comparison, and the other is identification, which is the process of one-to-many image matching and comparison.

Faces are characteristic, there are no two people in the world who are exactly the same, and the facial features of different individuals are also different, and each individual's face has its own unique facial features. The human eye observes the differences between these subtle features and transmits the observed facial features to our brain, which uses this information to make judgments about the identities of different individuals. The image acquisition device transmits the collected face image to the computer, the computer processes the received face image, models the face through the algorithm, and saves it in the computer, and makes a judgment on the identity of the same person when the computer receives the face image again .

With the development of face recognition technology to this day, its advantages cannot be ignored. First of all, face recognition technology is very convenient to use, the user only needs a camera or other image acquisition equipment, can complete the acquisition of the face, and the recognition process does not require the user to operate, which undoubtedly reduces the user's difficulty of use, and improves the user's sense of experience; Secondly, the security of face recognition technology is also very high, and face information is usually difficult to imitate, and users must perform face recognition by themselves to achieve the purpose of use, and it is difficult for others to imitate ; In the process of face image acquisition, the user does not have to contact the acquisition device, and can only keep the face acquisition intact at an appropriate distance. There are many advantages of face recognition technology, so I won't repeat them here.

In this paper, MATLAB is used to develop software to update the basic ORL face database, build a new face recognition library, use the PCA algorithm to reduce the dimension of face features, and pass the dimensionality-reduced face features into the SVM classifier for further processing. Training, establish a face recognition model based on the PCA-

SVM algorithm, and compare the recognition effect of the PCA model and the improved PCA-SVM model through examples to verify the reliability and accuracy of the model.

II. FACE DATABASE

As shown in Figure 1, the ORL face database contains 40 volunteers with different races[3], ages, and genders. Each person collected 10 images with different facial details, a total of 400 images, and the collection was completed. The image is then processed into a grayscale image.

Add new faces into the database

As shown in Figure 2, select 10 face images of people outside the ORL database, the image size is set to 92×112, and the image format is JPG, and added to the face database.

Figure 1 ORL face database.

Figure 2 Adding a new face to the face database.

III. PCA+SVM FACE RECOGNITION MODEL

A. PCA algorithm

Assuming that there is a set of linearly related m-dimensional features, through the PCA algorithm, the m-dimensional features are mapped to the k-dimensional space (where k is smaller than m) [4] to form a new set of k-dimensional orthogonal features, so as to achieve Dimensionality reduction for high-dimensional features. The process of PCA algorithm is shown in Fig.3.

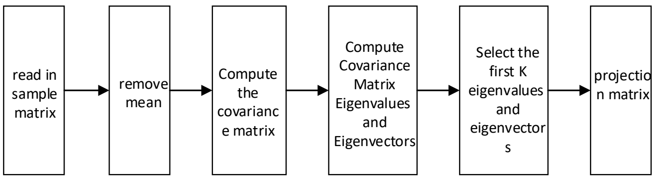

Figure 3 PCA flow chart.

(1) Data standardization processing

First perform data standardization processing, calculate the data set X by column, calculate the mean value Xm, and then normalize X_n=X- $X_m X_n$=X-X_m.

(2) Solve the covariance matrix of matrix Xn.

$$\sum = \frac{1}{N}\sum_{i=1}^{N}(X_i-\overline{X})(X_i-\overline{X})^T \tag{1}$$

(3) Calculate the eigenvalues and corresponding eigenvectors of the covariance matrix

$$\mathrm{cov}(X,Y)=\frac{\sum_{i=1}^{n}(X_i-\overline{X})(Y_i-\overline{Y})}{n-1} \tag{2}$$

$$F(p_1)=p_1E\{AA\}p_1^T-\lambda(p_1p_1^T-1) \tag{3}$$

(4) Sort the eigenvalues from large to small, select the largest k among them, and then use the corresponding k eigenvectors as column vectors to form an eigenvector matrix.

(5) Calculate the dimensionality-reduced data set, that is, project the normalized data set onto the selected eigenvector matrix, thus obtaining the dimension-reduced data set we need.

The facial features of a group of face images are often high-dimensional, because the elements contained in this group of images may not only include a face, but also various other elements, such as the sky, plants, various objects, etc., and the different environments in which the pictures are taken will also lead to different image backgrounds. In this case, face recognition is required, and the first premise is to erase the attributes that are not related to the face in the image, and only retain the features related to the face, which is the process of dimensionality reduction .

Through PCA dimensionality reduction, a set of low-dimensional facial features can be obtained. In face recognition technology, the PCA algorithm can be used to reduce the dimensionality of high-dimensional features of a face image and become low-dimensional features, thereby generating a new set of low-dimensional face feature images.

Because the system is based on the ORL face database, the ORL face database was selected to test the face features after PCA dimensionality reduction. Two sets of test data were constructed, and the recognition efficiency after dimensionality reduction was expressed from two aspects: the number of face samples and the dimensionality of face features, and the first 7 images of each group of samples were selected as the training set, and the last 3 images were selected as the test set.

B. SVM algorithm

The SVM algorithm is a kind of classification algorithm. Its principle is to map a set of high-dimensional data into points in space and then find a division hyperplane that forms the largest interval [5] (the reason for the largest interval is to ensure a low error rate). These points in the space can be well distinguished. The process of the SVM algorithm is as follows:

Define positive sample boundaries:

$$F(SV) = WTx + +b = 1 \tag{4}$$

Negative Sample Boundary:

$$F(SV) = WTx - +b = -1 \tag{5}$$

Then the distance between the two boundaries is equal to the point on the positive boundary:

$$x^* + (WTx^* + b) = 1 \tag{6}$$

The distance to the negative boundary is:

$$\frac{W^T x^* + b + 1}{\|W\|} = \frac{2}{\|W\|} \tag{7}$$

Because this distance function is not smooth enough (at 0), we expect to maximize the value of the function, so we can invert the function, and then find the minimum value of the inverse function [6], but this inverse function also exists in 0 is not smooth enough. So we still need to transform.

The SVM algorithm finally needs to obtain a classification decision function, and the classification decision function of the linearly separable support vector machine is[7]:

$$F(x) = sign\left(\sum_{i=1}^{N} a_i^* y_i (x x_i)\right) + b^* \tag{8}$$

The classification decision function of linear support vector machine is:

$$F(x) = sign\left(\sum_{i=1}^{N} a_i^* y_i (x x_i) + b^*\right) \tag{9}$$

IV. EXAMPLE VERIFICATION

A. PCA model

Through PCA dimensionality reduction, the high-dimensional features of a group of face images can be reduced to form a new set of low-dimensional face feature images [8]. Select the face features after PCA dimension reduction to test, and express the recognition efficiency after dimension reduction from the number of selected face samples and the dimension of face features, and select the first 7 samples of each group The images are used as the training set, and the last 3 images are used as the test set.

The facial features of a group of face images are often high-dimensional, because the elements contained in this group of images may not only include a face, but also various other elements, such as the sky, plants, various objects, etc., and the different environments in which the pictures are taken will also lead to different image backgrounds. In this case, face recognition is required, and the first premise is to erase the attributes that are not related to the face in the image, and only retain the features related to the face, which is the process of dimensionality reduction。

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During testing, the test set is grouped.

The number of samples selected for the first group is 20, and the number of samples selected for the second group is 40, that is, 20 and 40 different objects are selected in the face database as the samples of the test set, and the parameters after dimensionality reduction are respectively controlled by setting parameters. The face features are 10, 20, 30, 40, 60. For different feature dimensions, the recognition efficiency is evaluated from two aspects: recognition accuracy and recognition time. The results are shown in Table 1 and Table 2.

Table 1 PCA recognition effect when the sample size is 20.

Dimensionality of eigenface space	Recognition rate(%)	Recognition time (s)

10	79	0.27
20	85	0.31
30	86	0.33
40	87	0.35
60	88	0.40

Table 2 PCA recognition effect when the sample size is 40.

Dimensionality of eigenface space	Recognition rate(%)	Recognition time (s)
10	73	0.31
20	75	0.33
30	78	0.35
40	79	0.37
60	80	0.39

B. PCA+SVM model

The face recognition method based on PCA+SVM uses the support vector machine classifier to classify and recognize the face features after PCA dimensionality reduction[9]. Select the same experimental data as the PCA model, control different feature dimensions, and evaluate the recognition efficiency from two aspects: recognition accuracy and recognition time. The results are shown in Table 3 and Table 4.

Table3 PCA+SVM recognition effect when the sample size is 20.

Dimensionality of eigenface space	Recognition rate(%)	Recognition time (s)
10	90	0.24
20	92	0.26
30	94	0.27
40	95	0.28
60	96	0.30

Table4 PCA+SVM recognition effect when the sample size is 40.

Dimensionality of eigenface space	Recognition rate(%)	Recognition time (s)
10	88	0.34
20	82	0.36
30	89	0.37
40	90	0.38
60	90	0.39

C. Result analysis

It can be seen intuitively from Table 3 and Table 4 that when the number of samples is selected as 20, as the feature dimension increases, the recognition accuracy is not greatly affected, and the recognition time increases with the feature dimension. It can be seen that when the number of samples is low[10], the main factor affecting the recognition efficiency is the dimension of the feature.

Referring to Table 1 and Table 2, when the face feature dimension is low, the face recognition efficiency of the PCA+SVM model is about 20% higher than that of the PCA model, and with the continuous increase of the feature dimension, the PCA+SVM model The recognition time of the model is also lower than that of the PCA model. SVM algorithm is a type of classification algorithm, which works by mapping a set of high-dimensional data into points in space and finding a partition hyperplane that forms the maximum interval (the reason for the maximum interval is to ensure a low error rate), so that these points in space can be well distinguished.

Define positive sample boundary and negative sample boundary. So the distance between the two boundaries is equal to the distance from the point on the positive boundary to the negative boundary.

Because this distance function is not smooth enough (at 0), and we expect to maximize the function value, we can find the inverse function of the function and then find the minimum value of the inverse function . However, this inverse function also has the problem of not being smooth enough at 0. So further transformation is needed to obtain the function that satisfies the conditions.

V. CONCLUSION

In order to improve the application efficiency of face recognition technology, this paper proposes a face recognition model based on PCA-SVM. Firstly, the basic ORL face database is updated to build a new face recognition database; secondly, the PCA algorithm is used to reduce the dimensionality of face features, and the dimensionality-reduced face features are passed into the SVM classifier for training; finally, based on The PCA-SVM algorithm establishes a face recognition model, and compares the recognition effect of the PCA model and the improved PCA-SVM model through examples. The experiment proves that the face recognition accuracy of the PCA+SVM model is about 20% higher than that of the PCA model, and the recognition rate It is also more efficient. Therefore, the PCA+SVM proposed in this paper to establish a face recognition model has certain practical significance and can provide a reference for future exploration and research in the field of face recognition.

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